

GUI-SAGE: Enhancing GUI Automation with Self-Explanatory Learning

Fei Tang^{1,2*}, Zhangxuan Gu², Zhengxi Lu¹, Shangzhan Zhang¹, Zhengwen Zeng², Shuheng Shen^{2†},
Changhua Meng², Yuchen Yan¹, Wenqi Zhang¹, Yongliang Shen^{1‡}, Weiming Lu¹, Yueting Zhuang¹
¹Zhejiang University ²Ant Group

Abstract

Reinforcement learning with verifiable rewards (RLVR) has shown promise for GUI automation, enabling agents to learn from binary task completion signals. However, when task difficulty exceeds model capacity, on-policy exploration fails to discover correct actions, creating zero-advantage traps that eliminate learning signals. While incorporating off-policy expert demonstrations seems intuitive, it causes persistent high-entropy states due to distribution mismatch, disrupting effective learning. We propose GUI-SAGE, a self-explanation framework that generates in-distribution reasoning trajectories for GUI automation. By conditioning on ground-truth actions, our method produces in-distribution guidance that avoids the confusion caused by out-of-distribution expert demonstrations. We further introduce Entropy-Modulated Credit Assignment, which recalibrates learning weights by jointly considering prediction confidence and reward signals, enabling amplified updates for confident correct actions and attenuated updates for uncertain explorations. Extensive experiments on AndroidControl and GUI-Odyssey demonstrate that GUI-SAGE-3B achieves competitive performance with 81.1% success rate, substantially outperforming existing methods. Our analysis validates that self-explanations maintain stable learning dynamics while expert demonstrations cause entropy collapse, and that entropy modulation provides the largest improvements on in-distribution samples.

1. Introduction

Recent advances in Multimodal Large Language Models (MLLMs) [3, 4, 6, 17] have enabled significant progress in GUI automation, training agents to interact with digital interfaces through multi-step decision-making [10, 35, 39, 43]. Following the success of reinforcement learning with verifiable rewards (RLVR) in mathematical reasoning [8, 33], recent work has adopted this paradigm for

GUI tasks [25, 26, 46], where agents learn from binary task completion signals without requiring dense human annotations.

However, GUI automation faces a challenge that limits on-policy RLVR: *when task difficulty exceeds model capacity, exploration fails to discover correct actions*. The combinatorial nature of GUI tasks requires precise coordinate selection from high-resolution screens, appropriate action type choices, and correct text inputs, creating an enormous search space where random exploration rarely succeeds. This leads to what we term the *zero-advantage trap*: when all sampled rollouts fail, every trajectory receives identical zero rewards, producing uniform advantages that eliminate learning signals [33, 37, 46]. Our experiments reveal that **73.2%** of GUI tasks fall into this trap during early training (Table 5), fundamentally blocking learning progress.

The natural solution is incorporating expert demonstrations from more capable models to provide high-quality guidance when self-exploration proves insufficient. This approach has shown promise in other domains [44, 47], where off-policy data accelerates convergence and improves exploration efficiency. However, our investigation reveals that expert demonstrations actively harm learning in GUI automation. The core issue is **distribution mismatch**: expert reasoning from significantly more capable models (e.g., Gemini 2.5 Pro [6]) lies far outside the current policy’s distributional support. When we incorporate expert trajectories, the policy assigns extremely low log-probability to expert tokens (Figure 3a) and exhibits severe entropy increase (Figure 3c). This distributional incompatibility disrupts training dynamics: the model oscillates between attempting to mimic incomprehensible patterns and maintaining its original behavior, failing to learn actionable knowledge. Off-distribution experience cannot be effectively internalized due to the cognitive gap preventing meaningful learning.

This failure reveals a crucial principle for effective experience utilization: **learning signals must remain within the policy’s distributional support**. But how can we provide external knowledge while maintaining distribution compatibility? Our key insight is that *entropy can serve*

*This work was done when the first author was an intern at Ant Group.

†Project leader.

‡Corresponding author.

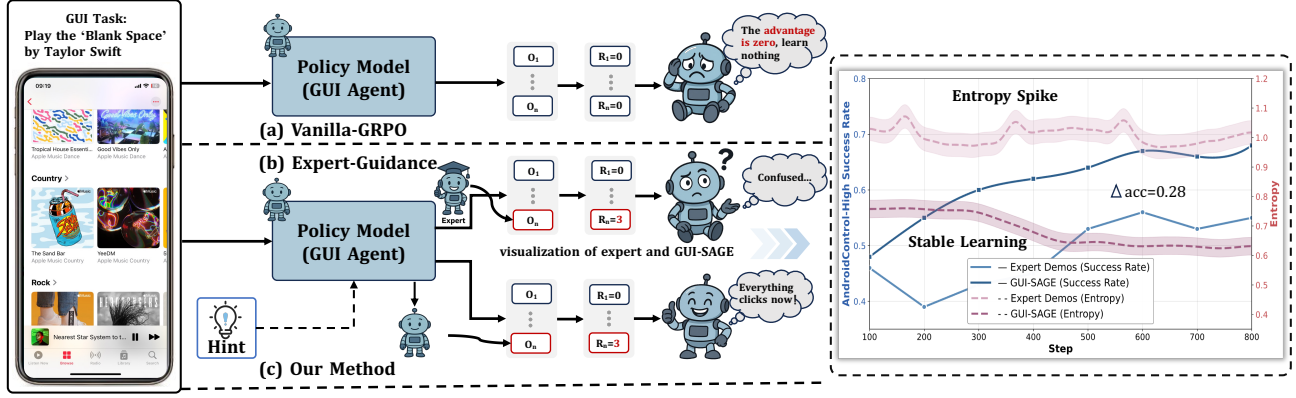


Figure 1. **Training approaches for GUI automation.** (a) Vanilla GRPO receive zero advantages when all rollouts fail. (b) Expert guidance causes distribution mismatch with high entropy (Figure 3c) and low log-probability (Figure 3a). (c) Self-explanation provides in-distribution guidance by conditioning on ground-truth actions. Right: Training dynamics showing success rate and entropy evolution across two methods, demonstrating that self-explanation maintains both high performance and stable entropy.

as a useful indicator of both distribution compatibility and prediction quality. When expert demonstrations are incorporated, entropy spikes reveal distribution mismatch. Within the policy’s own rollouts, entropy variance exposes confidence heterogeneity: confident predictions exhibit low entropy while uncertain explorations show high entropy, yet standard methods assign uniform credit regardless of this quality difference.

We introduce **GUI-SAGE**, a unified framework that addresses both challenges through entropy-aware learning. The core mechanism generates in-distribution guidance by conditioning the policy on ground-truth actions and allowing it to articulate reasoning in its own language. This self-explanation process produces trajectories using concepts the model already understands, maintaining stable entropy comparable to on-policy rollouts rather than the chaotic spikes caused by expert demonstrations (Figure 1). Building on this stable entropy profile, we incorporate **Entropy-Modulated Credit Assignment (EMCA)** into the learning objective, which leverages entropy as a natural proxy for prediction confidence. EMCA dynamically adjusts learning weights through regularized advantage: amplifying signals for confident correct actions (low entropy) while attenuating uncertain explorations (high entropy). This unified entropy-aware approach enables the model to learn from distribution-compatible guidance while differentiating reliable knowledge from spurious patterns.

Extensive evaluation on AndroidControl [18] and GUI-Odyssey [24] benchmarks demonstrates that GUI-SAGE achieves 81.1%, substantially outperforming existing methods. Our 3B model surpasses InfiGUI-R1-3B [23] by 5.6% and exceeds the larger UI-Venus-Navi-7B [11] by 1.1%, establishing that effective experience utilization can overcome both model scale and architectural differences. In-depth analysis validates our core hypotheses (Figure 1

(Right)): (1) self-explanations maintain stable entropy dynamics while expert demonstrations cause collapse, confirming distribution compatibility is essential; (2) EMCA provides largest performance gains (+1.1%), demonstrating effective confidence-aware credit assignment.

Our contributions are threefold:

- We propose GUI-SAGE, a self-explanation framework that generates in-distribution reasoning trajectories by conditioning on ground-truth actions, avoiding the high-entropy confusion caused by out-of-distribution expert demonstrations.
- We introduce Entropy-Modulated Credit Assignment that dynamically adjusts learning weights based on prediction confidence and reward signals, enabling differentiated updates across diverse sample qualities.
- We demonstrate through extensive experiments that GUI-SAGE achieves 81.1% average success rate, establishing competitive results while using smaller models than competing methods.

2. Related Work

2.1. GUI Agent

Recent MLLMs [3, 16, 22, 41, 45] have enabled significant progress in GUI automation [1, 2, 9, 12–14, 20, 27–29, 38]. Supervised learning approaches achieve strong performance through large-scale demonstrations: SeeClick [5] improves grounding via pre-training, ShowUI [21] proposes lightweight architectures, and UGround [10], OS-ATLAS [42], Aguis [43], UI-TARS [30] demonstrate improvements through supervised fine-tuning. Prompt-based methods like Cradle [36], Mobile-Agent [40] and AppAgent [15, 19, 48] leverage structured workflows, but require extensive expert design [30]. Following reinforcement learning with verifiable rewards (RLVR) success in

mathematical reasoning [8, 33], UI-R1 [26], GUI-R1 [26] and InfiGUI-R1 [23] adopt this paradigm with rule-based rewards, while some works [37, 46] develop continuous reward mechanisms. However, on-policy methods face critical limitations when task difficulty exceeds model capacity: exploration fails to sample correct actions, creating zero-advantage regimes that eliminate learning signals [33]. Our work addresses this through in-distribution guidance that maintains stable learning dynamics.

2.2. Off-Policy Learning

Incorporating external demonstrations to accelerate RL has been extensively studied. DAgger [32] addresses distribution shift through iterative aggregation, while recent work explores combining off-policy demonstrations with on-policy RL: LUFFY [44] applies off-policy gradients to incorporate high-quality reasoning trajectories from stronger models, and ExGRPO [47] utilizes experience replay from the model’s own successful rollouts to enhance sample efficiency. However, these approaches assume external experience lies within compatible distributions. Our investigation reveals that expert demonstrations from significantly more capable models cause severe distribution mismatch in GUI automation, manifesting as entropy spikes and training instability. We instead generate in-distribution guidance through self-explanation conditioned on ground-truth actions, avoiding the confusion caused by out-of-distribution expert demonstrations.

3. Method

3.1. Problem Formulation

We formulate GUI automation as a sequential decision-making problem. Given a task description t and initial screen state s_0 , the agent generates a sequence of actions $\{a_1, a_2, \dots, a_T\}$ to complete the task. At each step i , the agent observes the current screenshot state s_i and samples an action a_i from policy $\pi_\theta(a_i|t, s_i, a_{<i})$, where θ denotes model parameters and $a_{<i}$ represents action history. The action space $\mathcal{A}$ encompasses coordinate-based operations (click, swipe, long_press) and text input (type), as shown in Table 1. Upon task completion or reaching maximum steps T , the agent receives a sparse binary reward $R \in \{0, 1\}$ indicating success or failure. Our goal is to learn an optimal policy π_θ^* that maximizes expected reward across the task distribution.

3.2. GUI-SAGE Framework

Building on the insight that in-distribution samples drive effective policy learning, we propose GUI-SAGE, a unified framework that addresses both zero-advantage traps and prediction confidence heterogeneity through entropy-aware learning (Figure 2). The framework consists of two inte-

Action Type	Description
click	Click point at coordinate (x, y) .
long_press	Press point at (x, y) for specified seconds.
swipe	Swipe from (x_1, y_1) to (x_2, y_2) .
type	Input text into activated input box.
system_button	Press system button.
open	Open an app on device.
wait	Wait specified seconds for change.
terminate	Terminate task and report completion.

Table 1. Action space in GUI automation. Each action is defined by its type and corresponding parameters.

grated components: self-explanation generation produces reasoning trajectories conditioned on ground-truth actions, maintaining distribution compatibility while providing reliable learning signals; entropy-modulated credit assignment dynamically calibrates update magnitudes based on prediction confidence, enabling differentiated learning across diverse sample qualities.

3.2.1. Self-Explanation Generation

Given a GUI task t at state s with ground-truth action a^* , we generate self-explanations by conditioning the model on a^* as additional context. The model samples a reasoning trajectory c and action a from:

$$(c, a) \sim \pi_\theta(c, a \mid t, s, a^*) \quad (1)$$

This conditioning mechanism shifts the learning objective from action discovery to action explanation: rather than exploring the vast action space to find correct behaviors, the model articulates why the provided ground-truth action solves the task. Critically, since a^* guides generation, the resulting trajectory is guaranteed to produce correct actions, ensuring non-zero rewards during training.

During policy optimization, we augment on-policy rollouts with self-explained samples. For each task, we collect N responses: $N - 1$ rollout trajectories $\{(c_j, a_j)\}_{j=1}^{N-1} \sim \pi_\theta(\cdot \mid t, s)$ and one self-explained trajectory $(c_{se}, a_{se}) \sim \pi_\theta(\cdot \mid t, s, a^*)$. This mixed sampling strategy addresses the zero-advantage problem: when rollout samples fail to discover correct actions, the self-explained sample provides reliable positive signals by construction. Moreover, self-explanation maintains distribution compatibility because the model generates reasoning using its own vocabulary and conceptual structures, avoiding the entropy spikes caused by off-distribution expert demonstrations.

3.2.2. Entropy-Modulated Credit Assignment

Standard reinforcement learning methods assign uniform credit to all samples with identical rewards, ignoring the substantial variance in prediction confidence across trajectories. GUI automation exhibits particularly severe heterogeneity: confident correct predictions, overconfident errors,

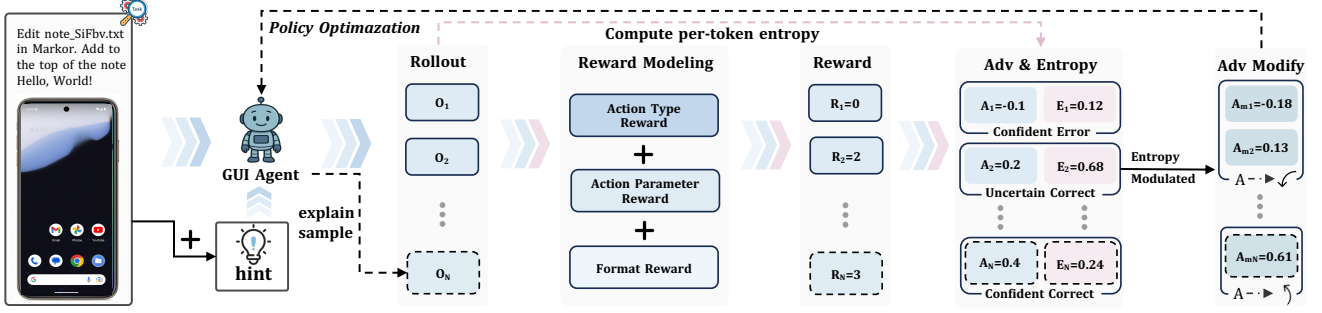


Figure 2. Overview of GUI-SAGE framework. Policy model generates in-distribution reasoning trajectories through self-explanation conditioned on hints. Entropy-Modulated Credit Assignment then recalibrates advantages based on prediction confidence (entropy), amplifying updates for confident correct predictions while attenuating uncertain explorations.

and uncertain explorations all receive equal treatment despite conveying fundamentally different information about policy quality. We propose Entropy-Modulated Credit Assignment (EMCA) that explicitly incorporates prediction confidence into advantage computation.

For each sampled trajectory, we compute per-token entropy during generation and obtain the average entropy H across the sequence. We first normalize entropy across all samples in the batch:

$$H_{\text{norm}} = \frac{H - \min(H)}{\max(H) - \min(H)} \quad (2)$$

Then apply exponential decay to obtain the entropy modulation factor:

$$g_H = \frac{\exp(-H_{\text{norm}})}{\mathbb{E}[\exp(-H_{\text{norm}})]} \quad (3)$$

where the expectation is computed over all samples in the current batch. The modulated advantage becomes:

$$A_{\text{mod}} = A \cdot g_H \quad (4)$$

where A is the original group-normalized advantage. This mechanism achieves fine-grained credit assignment: low-entropy predictions (whether correct or incorrect) receive amplified updates, enabling rapid consolidation of confident behaviors or strong penalties for systematic errors, while high-entropy explorations are attenuated to prevent noisy gradients from uncertain actions. Combined with self-explanation’s stable entropy profile, EMCA enables the model to differentiate reliable knowledge from potentially spurious patterns throughout training.

3.2.3. Reinforcement Learning with GUI-SAGE

We formulate the reward function to evaluate both action correctness and output structure. The total reward for each step comprises three components:

Action Type Reward. We define R_{type} to verify whether the predicted action type matches the ground-truth:

$$R_{\text{type}} = \begin{cases} 1, & \text{if action type matches,} \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Action Parameter Reward. Different action types require different parameter evaluations. For coordinate-based actions (e.g., click, swipe, long_press), we compute distance-based rewards; for text input actions (e.g., type), we use token-level F1-score.

Format Reward. To ensure structured output, we define R_{format} to verify that the model generates reasoning within `<think>` tags followed by action within `<tool.call>` tags:

$$R_{\text{format}} = \begin{cases} 1, & \text{if format is correct,} \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The final reward combines these components with weights w_1 and w_2 :

$$R = w_1 \cdot R_{\text{format}} + w_2 \cdot (R_{\text{type}} + R_{\text{param}}) \quad (7)$$

where w_1 controls the importance of structural correctness and w_2 governs action accuracy.

Policy Update. We integrate our GUI-SAGE framework into GRPO [33], which estimates advantages using multiple sampled responses. For each task with N samples (including $N - 1$ rollouts and 1 self-explanation), the advantage for sample i is computed as:

$$A_i = \frac{R(i) - \text{mean}(\{R(j)\}_{j=1}^N)}{\text{std}(\{R(j)\}_{j=1}^N)} \quad (8)$$

We then apply entropy-modulated credit assignment to obtain $A_{\text{mod},i}$. The policy optimization objective becomes:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta_{\text{old}}}} \left[\sum_t \min(r_t(\theta) A_{\text{mod},t}, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_{\text{mod},t}) - \beta D_{\text{KL}}[\pi_{\theta} \parallel \pi_{\text{ref}}] \right] \quad (9)$$

where $r_t(\theta) = \pi_{\theta}(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the importance sampling ratio, ϵ controls the trust region size, and β weights the KL regularization term that prevents excessive deviation from the reference policy. Figure 2 illustrates the complete framework. By combining self-explanation’s in-distribution guidance with entropy-aware credit assignment, GUI-SAGE enables effective learning even when on-policy exploration fails, addressing both zero-advantage traps and sample quality heterogeneity in GUI automation.

4. Experiment

4.1. Datasets and Implementation

Our training dataset contains approximately 40K samples from AndroidControl and GUI-Odyssey training sets. For evaluation, we use three representative GUI automation benchmarks: AndroidControl [18], GUI-Odyssey [24] and the challenging dynamics benchmark AndroidWorld [31], which assess action execution capabilities across diverse mobile applications.

We implement GUI-SAGE using Qwen2.5-VL [3] as the base model within the VLM-R1 framework [34]. Training is conducted on 8 NVIDIA A100-80G GPUs for 3 epoch with the following hyperparameters: learning rate 1e-6, train batch size 8, 8 sampled responses per instruction, and omit the KL penalty. For the reward weight, we set $w_1 = w_2 = 1.0$. We employ Flash Attention 2 [7] and use bfloat16 precision with gradient checkpointing. During inference, we use deterministic generation with temperature 0. For the evaluation, we followed Gu et al. [11].

4.2. Main Results

Table 2 shows that GUI-SAGE achieves competitive performance across two benchmarks. Starting from Qwen2.5-VL base models, our method substantially improves performance through self-explanation and entropy-modulated learning: GUI-SAGE-3B achieves 81.1% average success rate (SR), while GUI-SAGE-7B further improves to 82.1%.

Compared to baselines for SFT, GUI-SAGE-3B outperforms PAL-UI-3B by 26.1% in AndroidControl-High and 40.0% in GUI-Odyssey, with even greater gains over the 7B variant. Among RL methods, we surpass InfiGUI-R1-3B by 5.6% average SR, with particularly strong improvements in GUI-Odyssey (74.6% vs. 64.7%) and action type accuracy (92.1% vs. 81.6%). Our GUI-SAGE-3B also substantially

outperforms GUI-R1-7B (50.7% average SR) and SE-GUI-7B (52.8% on AndroidControl-High), validating that our self-explanation framework with entropy modulation achieves superior action prediction accuracy compared to existing RL approaches.

In particular, our 3B model remains competitive with larger models, surpassing AgentCPM-GUI-8B (78.1%) and matching UI-Venus-Navi-7B (80.0%). Our GUI-SAGE-7B model establishes new competitive results with 82.1% average SR, achieving 93.7% on AndroidControl-Low, 76.8% on AndroidControl-High, and 75.8% on GUI-Odyssey, demonstrating consistent improvements over the 3B variant across all benchmarks. These results validate that self-explanation learning with entropy modulation enables efficient policy optimization across model scales.

We additionally evaluate on AndroidWorld [31], a real-device benchmark with 116 tasks (Table 7), where GUI-SAGE-3B and -7B achieve 19.8% and 23.3% SR, substantially outperforming their Qwen2.5-VL base models.

5. Analysis

5.1. Analysis of Expert-CoT vs. Self-Explanation.

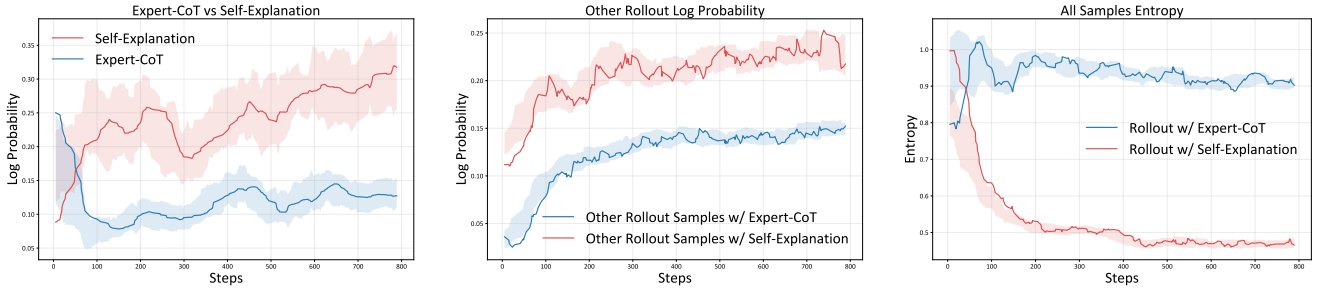
To validate our theoretical analysis, we use Qwen2.5-VL-72B [3] to generate expert trajectories and track log-probability and entropy during training under two augmentation strategies: expert-CoT versus self-explanations. As shown in Figure 3, we have the following key insights: (1) **Distribution shift**: Expert-CoT samples remain consistently out-of-distribution with significantly lower log-probability (Figure 3a). More critically, introducing a single expert-CoT sample substantially degrades other rollout samples’ log-probability (Figure 3b), reducing it to nearly half of that under self-explanations. (2) **Entropy dynamics**: Expert-CoT maintains persistently high entropy around 1.0 throughout training (Figure 3c), indicating persistent confusion. In contrast, self-explanations achieve rapidly decreasing entropy that stabilizes around 0.5, reflecting confident predictions without premature convergence. These results validate our theoretical analysis that in-distribution guidance enables effective policy learning.

5.2. Analysis of Training Dynamics.

To better understand the effectiveness of GUI-SAGE, we visualize training dynamics across 1,500 steps and compare with Vanilla-GRPO. As shown in Figure 4, we observe three key findings: (1) **Stable exploration**: Vanilla-GRPO exhibits rapid entropy collapse (dropping to 0.2), indicating premature convergence. Our method maintains stable entropy around 0.46, preserving necessary exploration while achieving confident predictions. (2) **Effective learning**: Our method achieves consistent response length (95 tokens) and consistently higher training rewards with more stable

GUI Agent Models	AndroidControl-Low		AndroidControl-High		GUI Odyssey		SR Avg
	Type acc	Step SR	Type acc	Step SR	Type acc	Step SR	
General Closed-source Models							
Claude	74.3	19.4	63.7	12.5	60.9	3.1	11.7
GPT-4o	74.3	19.4	66.3	20.8	34.3	3.3	14.5
GUI-specific SFT Models							
SeeClick	93.0	75.0	82.9	59.1	71.0	53.9	62.7
InternVL-2-4B	90.9	80.1	84.1	66.7	82.1	51.5	66.1
Aria-UI	-	67.3	-	10.2	-	36.5	38.0
OS-Atlas-7B	98.6	85.2	85.2	71.2	84.5	62.0	72.8
UI-TARS-2B	98.1	89.3	81.2	68.9	93.9	83.4	80.5
Aguvis-7B	-	80.5	-	61.5	-	-	-
Aguvis-72B	-	84.4	-	66.4	-	-	-
NaviMaster-7B	85.6	68.9	72.9	54.0	-	-	-
OS-Genesis-7B	90.7	74.2	66.2	44.5	-	-	-
SpiritSight-8B	-	87.6	-	68.1	-	75.8	77.2
PAL-UI-3B	-	-	60.4	49.3	56.7	34.6	-
PAL-UI-7B	-	-	71.3	57.8	65.1	41.7	-
GUI-specific RL Models							
InfGUI-R1-3B	94.8	91.1	81.8	70.7	81.6	64.7	75.5
UI-R1-3B	79.2	66.4	57.9	45.4	52.2	32.5	48.1
GUI-R1-3B	83.7	64.4	58.0	46.5	54.8	41.3	50.7
GUI-R1-7B	85.1	66.5	71.6	51.8	65.5	38.8	52.4
UI-AGILE-7B	87.7	77.6	80.1	60.6	-	-	-
SE-GUI-7B	-	68.2	-	52.8	-	-	-
UI-TARS-1.5-7B	95.3	87.0	77.4	62.3	-	-	-
AgentCPM-GUI-8B	94.4	90.2	77.7	69.2	90.0	75.0	78.1
UI-Venus-Navi-7B	97.1	92.4	86.5	76.1	87.3	71.5	80.0
Base Models							
Qwen2.5-VL-3B	-	90.8	-	63.7	-	-	-
Qwen2.5-VL-7B	-	91.4	-	60.1	-	-	-
Ours							
GUI-SAGE-3B	95.5	93.4	86.4	75.4	92.1	74.6	81.1
GUI-SAGE-7B	96.0	93.7	87.0	76.8	93.2	75.8	82.1

Table 2. We report high and low settings on AndroidControl. Step SR represents step success rate. Avg SR represents the average SR across the three benchmarks (AndroidControl-Low, AndroidControl-High, and GUI Odyssey). “-” indicates missing values due to unavailable results in the original paper, unreleased model checkpoints, and inference code.



(a) Log-probability of the augmented sample (expert-CoT or self-explanation).

(b) Log-probability of other rollout samples (excluding the augmented one).

(c) Entropy across all rollout samples.

Figure 3. Training dynamics comparison between expert-CoT and self-explanation augmentation. Expert demonstrations remain out-of-distribution, suppress rollout probabilities, and maintain high entropy, while self-explanations enable stable in-distribution learning with increasing confidence.

Methods	AC-Low SR	AC-High SR	Avg
Vanilla-GRPO	89.7	70.4	80.1
<i>Hint-based Baselines</i>			
ATH	90.8	71.9	81.4
APH	91.2	72.4	81.8
<i>Ours</i>			
Self-Explanation	91.9	74.2	83.1

Table 3. Ablation study on hint formats. Avg represents the average of two SR metrics. Self-explanation conditioned on complete ground-truth action outperforms all partial hint variants.

curves, while Vanilla-GRPO shows drastically shortened responses (from 130 to 50 tokens) and unstable reward dynamics. (3) **Sparse action learning**: Most critically, sparse action rewards at termination steadily improve from 1.5 to over 2.0, while Vanilla-GRPO stagnates around 1.0 throughout training. The termination action is particularly challenging as it requires the model to accurately assess task completion based on cumulative state changes. This demonstrates that self-explanation guidance enables effective learning from sparse signals, successfully discovering correct actions where vanilla GRPO fails to make progress.

5.3. Analysis of Hint Formats.

To understand what type of guidance most effectively improves policy learning, we investigate different hint formats. A complete ground-truth action contains both action type and action parameter. We compare four approaches: (i) **Vanilla-GRPO**: no hint, relying solely on exploration; (ii) **ATH (Action Type Hint)**: conditioning only on action type; (iii) **APH (Action Parameter Hint)**: conditioning only on action parameters; (iv) **Self-Explanation**: conditioning on the complete ground-truth action to generate reasoning trajectories. Note that all methods in this ablation use standard GRPO without entropy-modulated credit assignment to isolate the effect of different hint formats.

As shown in Table 3, all hint-based methods improve over Vanilla-GRPO, with performance gains correlating with the amount of provided information. ATH yields modest gains (+1.3% average), while APH achieves slightly better performance (+1.7% average). However, even with partial information, the model still faces significant challenges in discovering correct complete actions through exploration in the highly discrete and sparse action parameter space. In contrast, our self-explanation approach achieves the best performance (83.1% average). By conditioning on the complete ground-truth action, the model focuses entirely on generating coherent reasoning trajectories that explain why this action is correct. This complete guidance enables the model to achieve better performance across GUI automation tasks.

Sparse Action Type	Vanilla-GRPO (%)
long_press	91.3
terminate	87.6
system_button	84.2
Average Sparse Actions	87.7

Table 4. Zero-advantage rate (%) for sparse actions during early training (0-300 steps). Vanilla GRPO suffers from severe zero-advantage problem on sparse actions (87.7% average).

Training Steps	Zero-Adv (%)	Non-Zero-Adv (%)
0-100	73.2	26.8
100-200	69.5	30.5
200-300	68.9	31.1
Overall (0-300)	67.9	32.1

Table 5. Distribution of zero-advantage vs. non-zero-advantage samples across vanilla-GRPO training steps.

w_1	w_2	AC-Low		AC-High		Avg
		Type	SR	Type	SR	
0.2	0.8	96.0	93.1	85.0	74.9	83.5
1.0	1.0	96.2	93.4	85.3	75.4	84.2
0.8	0.2	95.9	92.9	84.8	74.6	83.7

Table 6. Ablation study on reward weights w_1 (format reward) and w_2 (action reward). Equal weighting achieves the best performance on GUI-SAGE-3B.

5.4. Analysis of Zero-Advantage in Early Training

To analyze the zero-advantage problem in vanilla GRPO training, we visualize rollout samples during early training (0-300 steps). Table 5 shows that 73.2% of samples in the initial phase (0-100 steps) receive uniform zero rewards, providing no learning signal. This problem persists throughout training, with 67.9% overall samples suffering from zero advantages. The problem is particularly severe for sparse actions. As shown in Table 4, 91.3% of long_press and 87.6% of terminate actions fall into the zero-advantage trap, as these actions are rarely sampled during rollouts. On average, 87.7% of sparse action samples provide no learning signal, fundamentally blocking the model’s ability to discover correct behaviors for these critical actions. Our GUI-SAGE method addresses this limitation by providing in-distribution guidance through self-explanations conditioned on ground-truth actions, ensuring effective learning signals even for sparse actions.

5.5. Analysis of Reward Weight

We investigate the impact of reward weights w_1 and w_2 in Equation 7, which balance format correctness and action accuracy. Table 6 shows that equal weighting ($w_1 = w_2 = 1.0$) achieves optimal performance with 84.2% average SR. Overemphasizing format ($w_1 = 0.8, w_2 = 0.2$) leads to a slight performance drop to 83.7%, as excessive focus

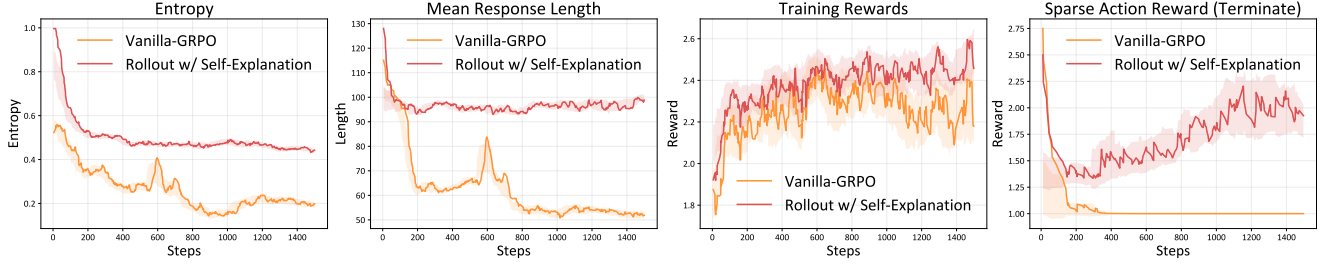


Figure 4. Training dynamics comparison. Our method achieves stable entropy, longer responses, higher training rewards, and significantly better sparse action rewards at termination compared to Vanilla-GRPO.

Model	SR (%)
Qwen2.5-VL-3B	3.5
Qwen2.5-VL-7B	19.0
GUI-SAGE-3B	19.8
GUI-SAGE-7B	23.3

Table 7. Performance on AndroidWorld benchmark. SR represents success rate across 116 dynamically instantiated tasks.

on structural correctness may constrain action exploration. Similarly, underweighting format ($w_1 = 0.2, w_2 = 0.8$) results in 83.5%, indicating that proper output structure remains important for effective learning. The relatively stable performance across different weight configurations (± 0.5 - 0.7%) demonstrates that our method is robust to reward weighting, while equal weighting provides the optimal balance between format correctness and action accuracy.

5.6. Ablation Study

To comprehensively evaluate our approach, we compare four training strategies: (1) **Vanilla-GRPO w/o Reasoning**: directly output action prediction without reasoning generation, (2) **Vanilla-GRPO**: generating reasoning before action prediction, (3) **Expert Demonstration**: augmenting with trajectories from Qwen2.5-VL-72B, and (4) **Self-Explanation**: our method generating in-distribution reasoning. We further investigate the effectiveness of Entropy-Modulated Credit Assignment across all methods. As shown in Table 8, we have the following key insights: (1) **Reasoning is beneficial for GUI automation**. GRPO w/o Reasoning achieves 78.5% average SR (79.3%/77.7% with/without entropy modulation), substantially lower than Vanilla-GRPO’s 80.1%. This demonstrates that explicit reasoning generation before action prediction provides significant benefits for GUI navigation tasks, enabling the model to develop structured decision-making processes. (2) **Self-explanation outperforms expert guidance**. Expert-CoT reaches 81.2% but struggles with distribution mismatch (Figure 3a). Our self-explanation approach achieves 83.1% average SR, outperforming all baselines by at least 1.9%, validating that in-distribution guidance is more ef-

Methods	AC-Low		AC-High		Avg
	Type	SR	Type	SR	
Vanilla-GRPO w/o Reasoning	94.8	88.2	81.5	68.8	78.5
w/ EMCA	95.1	88.6	82.0	69.3	79.0
Vanilla-GRPO	95.5	89.7	83.4	70.4	80.1
w/ EMCA	95.8	90.2	83.9	70.9	80.6
Expert Demonstration	96.1	90.7	83.2	71.6	81.2
w/ EMCA	96.4	91.2	83.7	72.3	81.9
<i>Ours</i>					
Self-Explanation	95.6	91.9	84.6	74.2	83.1
w/ EMCA	96.2	93.4	85.3	75.4	84.2

Table 8. Ablation study comparing training strategies. Avg represents the average of two SR metrics.

fective than out-of-distribution expert demonstrations. (3) **EMCA amplifies in-distribution learning**. The EMCA component consistently benefits all methods (+0.4% for GRPO w/o Reasoning, +0.5% for Vanilla-GRPO, +0.7% for Expert-CoT). Notably, it achieves the largest improvement (+1.1%) when combined with self-explanation, reaching 84.2% average SR. This larger gain likely occurs because self-explanations are in-distribution samples that the model better understands, leading to higher learning weights and more effective policy updates.

6. Conclusion

In this paper, we propose GUI-SAGE, a self-explanatory learning framework that addresses the challenge of effective reinforcement learning in GUI automation. Unlike vanilla GRPO that suffers from zero advantages on hard tasks, our method generates reasoning trajectories by conditioning on ground-truth actions, providing in-distribution guidance that avoids the high-entropy confusion caused by expert demonstrations. We further introduce Entropy-Modulated Credit Assignment, which dynamically recalibrates learning weights by jointly considering prediction confidence and reward signals, enabling differentiated updates across diverse sample qualities. Extensive experiments demonstrate that GUI-SAGE-3B achieves 81.1% average success rate on AndroidControl and GUI-Odyssey benchmarks, outperforming AgentCPM-GUI-8B by 3.0%.

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